

The Newsvendor and its variants



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Managerial motivation

Introduction: Managerial motivation

- ▶ The Newsvendor (“newspaper seller”): choosing a quantity *before* observing demand
- ▶ Class of the problem: **1D convex / scenario-based stochastic LP**
- ▶ It is the gateway to **stochastic optimization**

A very common decision

- ▶ Ordering too much generates **unsold items**; too little, **lost sales**
- ▶ It applies to fashion and seasonal goods, fresh and perishable products, drugs, hotel capacity, spare parts
- ▶ Its didactic strength is a clear-cut result:

the optimal quantity is not the average demand

We decide first and find out later

- ▶ The model is more than a century old and remains the foundation of inventory management for single-cycle products
- ▶ It is the first encounter with the idea that *we decide first and find out later*
- ▶ And with the tools needed to do it well: scenarios, value of the stochastic solution, risk measures

In this chapter we will learn to

1. derive and apply the quantile rule
2. turn a stochastic problem into a scenario-based LP with the positive-part trick
3. quantify the value of modelling uncertainty (VSS)
4. incorporate service levels and risk aversion



Summary: Managerial motivation

- ▶ A single quantity chosen before knowing demand
- ▶ Two possible errors: unsold items and lost sales
- ▶ The optimal quantity is **not** the average demand

The basic model

Introduction: The basic model

- ▶ We formulate the Newsvendor as the minimization of an expected cost
- ▶ A single variable: the order quantity q
- ▶ We derive the **quantile rule**, which gives the optimum in closed form

The problem in words

- ▶ *We decide* a single quantity q , before knowing demand
- ▶ *The objective* is to minimize the total expected cost of the error: having ordered too much (unsold items) plus having ordered too little (lost sales)
- ▶ *Constraints* are absent in the basic model ($q \geq 0$)
- ▶ They appear in the variants: shared budget, service levels, capacity limits

Data (input of the model) (1/2)

Symbol	Type	Meaning
D	random var. ≥ 0	demand, with cumulative distribution function F
c_u	$\in \mathbb{Q}_{>0}$	unit <i>underage</i> cost (unmet demand): lost margin plus any penalty (€)
c_o	$\in \mathbb{Q}_{>0}$	unit <i>overage</i> cost (unsold unit): cost minus salvage value (€)

Data (input of the model) (2/2)

- ▶ D is an exception to the convention on capital letters, to follow the literature: the random variable in upper case, its realizations in lower case (d_s)
- ▶ With selling price p , purchase cost c and salvage value $v \leq c$ (all in $\mathbb{Q}_{>0}$):

$$c_u = p - c$$

$$c_o = c - v$$

Decision variable and model

We introduce a non-negative variable:

q = quantity ordered *before* observing demand D

Definition 1: Newsvendor model in cost form

$$\min \quad c_o \mathbb{E}[(q - D)^+] + c_u \mathbb{E}[(D - q)^+]$$

subject to $q \geq 0$

Objective function and constraint

- ▶ The objective function is **convex** and minimizes the expected cost of the error:
 - ▶ overage $(q - D)^+$ charged at c_o
 - ▶ underage $(D - q)^+$ charged at c_u
- ▶ The symbol $(x)^+ = \max\{x, 0\}$ denotes the positive part
- ▶ The constraint $q \geq 0$ defines the variable of the model

The quantile rule

If F is continuous, the optimum satisfies the **quantile rule**:

$$F(q^*) = \alpha^* = \frac{c_u}{c_u + c_o} \quad \Rightarrow \quad \boxed{q^* = F^{-1}(\alpha^*)}$$

Where the quantile rule comes from

- ▶ The argument works on the marginal unit: ordering the q -th extra unit
 - ▶ yields c_u if demand absorbs it (probability $1 - F(q)$)
 - ▶ costs c_o if it remains unsold (probability $F(q)$)
- ▶ It is worth doing as long as $c_u(1 - F(q)) \geq c_o F(q)$, that is, as long as

$$F(q) \leq \frac{c_u}{c_u + c_o}$$

- ▶ Underage far more expensive than overage: α^* is high, one orders a high quantile
- ▶ Unsold items expensive (perishables): one orders little

Summary: The basic model

- ▶ Expected cost $c_o \mathbb{E}[(q - D)^+] + c_u \mathbb{E}[(D - q)^+]$, convex in q
- ▶ $c_u = p - c$ (underage), $c_o = c - v$ (overage)
- ▶ Quantile rule: $q^* = F^{-1}(c_u/(c_u + c_o))$

Worked example by hand (discrete demand)

Introduction: Worked example by hand

- ▶ Three equally likely demand scenarios
- ▶ We compare the expected cost of the natural candidates
- ▶ We check that the quantile rule (discrete version) gives the same answer

Example: three equally likely scenarios (1/3)

Data: $D \in \{80, 100, 120\}$ with probability $1/3$; $c_u = 9$, $c_o = 4$, hence

$$\alpha^* = 9/13 = 0.6923$$

Step 1 — expected cost for the three natural candidates.

$$C(80) = \frac{1}{3} [0 + 9 \cdot 20 + 9 \cdot 40] = 180$$

$$C(100) = \frac{1}{3} [4 \cdot 20 + 0 + 9 \cdot 20] = \frac{260}{3} = 86.7$$

$$C(120) = \frac{1}{3} [4 \cdot 40 + 4 \cdot 20 + 0] = 80$$

Example: three equally likely scenarios (2/3)

Step 2 — discrete quantile rule.

- ▶ We need the smallest q with $F(q) \geq 0.6923$:

$$F(80) = 1/3$$

$$F(100) = 2/3 = 0.667 < 0.6923$$

$$F(120) = 1$$

- ▶ Hence $q^* = 120$: consistent with step 1

Example: three equally likely scenarios (3/3)

Step 3 — interpretation.

- ▶ Ordering the historical *maximum* even though the mean is 100
- ▶ With $c_u = 9 \gg c_o = 4$, running short costs more than twice as much as running long
- ▶ Whoever orders “the mean” pays 86.7 against 80
- ▶ Uncertainty is managed, not ignored

Summary: Worked example by hand

- ▶ With $\alpha^* = 0.6923$ the optimal quantity is $q^* = 120$, the historical maximum
- ▶ Ordering the mean costs 86.7 against 80
- ▶ Discrete quantile rule: the smallest q with $F(q) \geq \alpha^*$

Case study: the quantile rule in the continuous case

Introduction: the quantile rule in the continuous case

- ▶ A bakery: $p = 15$, $c = 6$, $v = 2 \text{ €} \Rightarrow c_u = 9$, $c_o = 4$
- ▶ Normally distributed demand with $\mu = 100$, $\sigma = 20$
- ▶ We apply the quantile rule and see how stable the solution estimated from scenarios is

Results

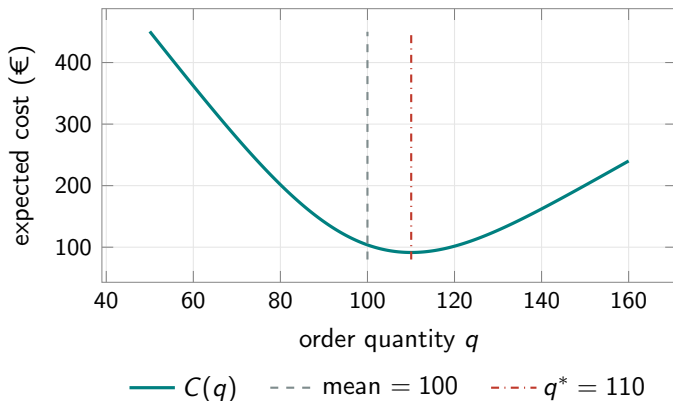
Bakery: $p = 15$, $c = 6$, $v = 2 \text{ €} \Rightarrow c_u = 9$, $c_o = 4$; normally distributed demand with $\mu = 100$, $\sigma = 20$.

$$\alpha^* = 9/13 = 0.6923$$

$$q^* = F^{-1}(0.6923) = 110.05 \text{ units} \quad (\text{mean} = 100)$$

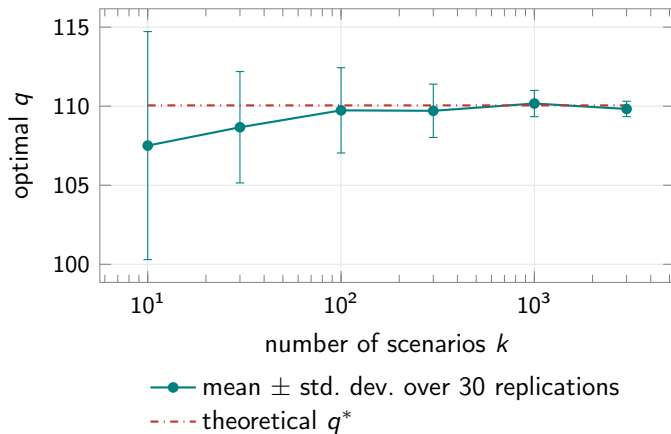
$$\text{expected cost at } q^*: 91.43 \text{ EUR} \quad \text{at } q = \text{mean}: 103.72 \text{ EUR}$$

Minimum at the 69th percentile



The expected cost is convex; the minimum falls at $q^* = 110$, not at the mean 100.

Stability of the scenario-based solution



Reading the figures

- ▶ The expected cost $C(q)$ is convex and the minimum falls at $q^* = 110$ (69th percentile of demand), not at the mean 100
- ▶ To the right of the mean one pays less: at q^* the expected cost is 91.43 € against 103.72 € at the mean
- ▶ The optimal q estimated from k sampled scenarios (mean and standard deviation over 30 replications) shows that with few scenarios the decision is **unstable**

Summary: the quantile rule in the continuous case

- ▶ $\alpha^* = 0.6923$, $q^* = 110.05$ against a mean of 100
- ▶ Ordering the mean costs 12.29 € more per cycle
- ▶ Solutions estimated from few scenarios swing a lot

The scenario-based linear formulation

Introduction: The scenario-based linear formulation

- ▶ When F is not known in closed form — the normal situation in a company — one uses **scenarios**
- ▶ Historical data, samples from an estimated distribution, or weighted scenarios
- ▶ We obtain an **LP** thanks to the positive-part trick
- ▶ We measure the value of the stochastic solution (VSS)

New data and variables

- ▶ Number of scenarios $k \in \mathbb{Z}_{\geq 1}$, indexed by $s \in \{1, 2, \dots, k\}$
- ▶ For each scenario: demand $d_s \in \mathbb{Q}_{\geq 0}$ and probability $\pi_s \in \mathbb{Q}_{\geq 0}$, with $\sum_{s=1}^k \pi_s = 1$
- ▶ Besides q , we introduce $2k$ non-negative variables:

$$\begin{cases} o_s = \text{overage (unsold units) in scenario } s \\ u_s = \text{underage (unmet demand) in scenario } s \end{cases}$$

for every $s \in \{1, 2, \dots, k\}$

Definition 2: Scenario-based Newsvendor LP model

$$\min \sum_{s=1}^k \pi_s (c_o o_s + c_u u_s)$$

$$\text{subject to } o_s \geq q - d_s, \quad \forall s \in \{1, \dots, k\}$$

$$u_s \geq d_s - q, \quad \forall s \in \{1, \dots, k\}$$

$$q \geq 0$$

$$o_s, u_s \geq 0, \quad \forall s \in \{1, \dots, k\}$$

Objective function and constraints

- ▶ The linear objective function is the weighted average, over the scenarios, of the overage and underage costs: the “scenario-based” version of the expected cost
- ▶ The constraints $o_s \geq q - d_s$ and $u_s \geq d_s - q$ define the overage and the underage of each scenario starting from the only genuine decision q
- ▶ Together with the non-negativity conditions they realize the positive parts ($2k$ linear constraints)
- ▶ $q \geq 0$ and $o_s, u_s \geq 0$ define the variables

Why it is linear: the positive-part trick

- ▶ $(q - d_s)^+$ is not linear
- ▶ At the *optimum* the variable o_s , which appears in the objective with a positive coefficient, is pushed down onto the maximum between $q - d_s$ and 0
- ▶ The constraint $o_s \geq q - d_s$ together with $o_s \geq 0$ does exactly the job of $\max\{q - d_s, 0\}$ without binary variables
- ▶ The same mechanism applies to u_s
- ▶ The trick returns unchanged in the **CVaR** and in the **SVM**

The model in Gurobi

```
q = m.addVar(name="q")
o = m.addVars(S, name="o")      # overage per scenario
u = m.addVars(S, name="u")      # underage per scenario
m.addConstrs((o[s] >= q - dom[s] for s in range(S)))
m.addConstrs((u[s] >= dom[s] - q for s in range(S)))
m.setObjective(gp.quicksum((Co*o[s] + Cu*u[s]) / S
                           for s in range(S)), GRB.MINIMIZE)
```

Results

LP with $k = 600$ scenarios: $q = 108.61$
empirical quantile at 69.23%: 108.60 (they coincide: it is a theorem)
VSS: decision $q = E[D] = 100$ costs 99.50
stochastic q costs 88.33
value of the stochastic solution = 11.17 EUR per cycle

Managerial insight (1/2)

- ▶ The *value of the stochastic solution* (VSS) answers the boss's question: "why complicate our lives with scenarios?"
- ▶ Here it is worth 11.17 € per cycle, 11% of the cost
- ▶ It is the gain from *modelling* uncertainty instead of replacing it with the mean

Managerial insight (2/2)

- ▶ The figure on stability gives the other half of the message
- ▶ Below 100 scenarios the optimal q swings by ± 3 units from one estimate to the next
- ▶ Scenarios are themselves a sample, and the decision inherits their variance

Summary: The scenario-based linear formulation

- ▶ With k scenarios the Newsvendor becomes an LP with $2k + 1$ variables
- ▶ The positive parts are linearized without binary variables
- ▶ The LP reproduces the empirical quantile (it is a theorem)
- ▶ $VSS = 11.17$ € per cycle: modelling uncertainty pays off

Service levels

Introduction: Service levels

- ▶ Contracts often impose a **service level**
- ▶ *Cycle service level*: probability of not running into a stock-out
- ▶ *Fill rate*: share of demand served on average
- ▶ We see how much they cost and why they must not be confused

The cost of service levels

```
cycle service level 90%: q = 125.63
                        (cost +23.41 EUR over the economic
                        optimum)
cycle service level 95%: q = 132.90 (cost +45.59)
cycle service level 99%: q = 146.53 (cost +95.56)
fill rate at q*: 96.06%
probability of no stockout at q*: 69.23%
```

Do not confuse the service levels

Observation 1: Do not confuse the service levels

At $q^ = 110$ the probability of avoiding a stock-out is only 69%, but the fill rate (share of demand served on average) is 96%: different measures that contracts often confuse.*

- ▶ Promising “95% probability of full coverage” costs +45.59 € per cycle compared with the economic optimum
- ▶ Promising “95% fill rate” is almost free
- ▶ Read the SLA carefully before signing it

Summary: Service levels

- ▶ Raising the cycle service level costs more and more: +23, +46, +96 € for 90%, 95%, 99%
- ▶ At q^* : no stockout 69%, but fill rate 96%
- ▶ Two different measures: read the SLA carefully

Risk aversion: mean-CVaR

Introduction: Risk aversion

- ▶ Expected cost does not distinguish between many small losses and rare disasters
- ▶ With the linear formulation of the CVaR (here only anticipated) we minimize

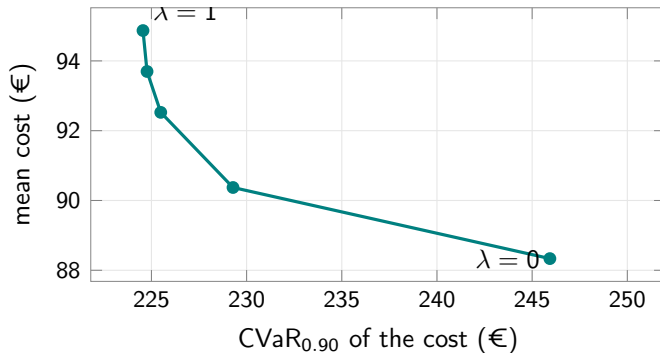
$$(1 - \lambda) \mathbb{E}[\text{cost}] + \lambda \text{CVaR}_{0.90}(\text{cost})$$

- ▶ The weight $\lambda \in [0, 1]$ measures risk aversion

Results as λ varies

lambda = 0.00:	q = 108.61	mean cost	88.33	CVaR90	245.93
lambda = 0.25:	q = 112.45	mean cost	90.37	CVaR90	229.29
lambda = 0.50:	q = 114.48	mean cost	92.52	CVaR90	225.49
lambda = 1.00:	q = 116.17	mean cost	94.87	CVaR90	224.56

Mean cost – tail risk frontier



Reading the frontier

- ▶ From $\lambda = 0$ (optimal on average, heavy tail) to $\lambda = 1$ (maximum protection)
- ▶ The first stretch is almost vertical: a lot of protection for very little mean cost
- ▶ The risk-averse decision maker orders more (from 108.6 to 116.2 units)
- ▶ Pays 6.5 € more on average in order to cut by 21.4 € the mean cost of the worst 10% of the scenarios
- ▶ Most of the protection is already obtained with $\lambda = 0.25$

Summary: Risk aversion

- ▶ Mean-CVaR objective: a trade-off between expected cost and tail risk
- ▶ More risk aversion $\Rightarrow$ one orders more
- ▶ The frontier is steep at the beginning and flat afterwards: a little aversion gives a lot of protection

Multi-product with a shared budget

Introduction: Multi-product with a shared budget

- ▶ Three cakes with *correlated* demands ($\rho = 0.7$): the holidays go well or badly for everyone together
- ▶ A shared production budget of 1200 €
- ▶ We see how the budget compresses the quantities and what the dual says

Results

product	q without budget	q with budget
panettone	111.1	99.3
pandoro	91.5	75.4
torrone	65.9	56.8
Spend: 1200/1200 shadow price of the budget: -0.553		

Reading the results (1/2)

- ▶ The binding budget compresses the three quantities below their respective optimal quantiles
- ▶ But not proportionally: the cut depends on the ratio c_u/c_o and on the unit cost of each product
- ▶ The dual says that one more euro of budget would reduce the expected cost by 0.55 €

Reading the results (2/2)

- ▶ A 55% return that would justify almost any financing
- ▶ Information that nobody would have quantified without the model
- ▶ The high correlation removes the diversification benefit: the three products fail together
- ▶ Treating them as independent would underestimate the risk

Summary: Multi-product with a shared budget

- ▶ A shared budget cuts the quantities in a non-proportional way
- ▶ The shadow price (-0.553) measures the value of one more euro of budget
- ▶ With correlated demands there is no diversification

Exercises

Introduction: Exercises

- ▶ Six exercises to consolidate the concepts of the chapter
- ▶ Quantile rule, historical data, perishables, EVPI, fill rate, multi-product
- ▶ They are solved by hand and with the laboratory script

Exercise: check

- ▶ The municipality imposes a penalty $b = 3 \text{ €}$ per unserved customer:

$$c_u = p - c + b$$

- ▶ Recompute by hand α^* and q^*
- ▶ Check with the script

Exercise: historical data

- ▶ Use as scenarios the 24 demand observations of the last two years (generate them with the script) instead of the distribution
- ▶ How much does q differ?
- ▶ Which approach would you recommend to a company that has only 24 observations?

Exercise: perishables

- ▶ The salvage value becomes negative: $v = -1$ € (disposal costs money)
- ▶ How do α^* and q^* change?
- ▶ Plot q^* as a function of $v \in [-2, 4]$

Exercise: EVPI

- ▶ Compute the EVPI (*Expected Value of Perfect Information*)
- ▶ Mean cost with q chosen *after* having seen demand (that is, $q = d_s$, cost 0) against the cost of the stochastic solution
- ▶ How much is a perfect demand forecast worth at most?

Exercise: constrained fill rate

- ▶ Add to the LP the fill rate constraint

$$\sum_{s=1}^k \pi_s u_s \leq (1 - 0.98) \mathbb{E}[D]$$

- ▶ Compare quantity and cost with the cycle service level at 98%

Exercise: multi-product

- ▶ Rerun the multi-product case with independent demands ($\rho = 0$), everything else being equal
- ▶ Does the optimal expected cost improve or worsen?
- ▶ Why?

Summary (1/2)

Basic model:

- ▶ We decide q before observing demand
- ▶ Quantile rule: $q^* = F^{-1}(c_u/(c_u + c_o))$, not the mean

Scenario-based formulation:

- ▶ LP with overage and underage variables (positive-part trick)
- ▶ VSS: the gain from modelling uncertainty
- ▶ Few scenarios $\Rightarrow$ unstable decision

Summary (2/2)

Variants:

- ▶ Service levels: cycle service level and fill rate are different measures
- ▶ Mean-CVaR: the risk-averse decision maker orders more; a little aversion gives a lot of protection
- ▶ Shared budget: the shadow price measures the value of one more euro